

A Calibrated Separation of the Kontorovich–Lagarias and Volkov Exponent Predictions for the $5x + 1$ Reverse Tree

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August 10, 2026

Abstract

Kontorovich and Lagarias predict a counting exponent $\eta \approx 0.650919$ for the reverse tree of the accelerated $5x + 1$ map, where $N_u(x) = x^{\eta+o(1)}$. A competing model due to Volkov predicts $\eta^* \approx 0.678$. The two authors write that Volkov’s own data cannot discriminate between the values.

We enumerate the reverse tree exactly and extrapolate the truncation bias with Aitken’s Δ^2 . The fixed-window reading is 0.639; it excludes neither prediction. Applied to a branching process of known exponent, the same estimator under-reads by 0.038, more than the gap $\Delta = 0.027$ between the two disputed values, so a raw reading settles nothing.

We build synthetic controls that share the arithmetic tree’s branching law: two also share, via an exact sibling congruence, its sibling spacing, drawing the branch residue instead of the true one; a third draws every branch label independently, with no such congruence. Three such controls of exponent 0.650919 read within 0.0035 of that value and within 0.004 of each other. The arithmetic tree reads 0.64926, below the two congruence-matched controls and inside the wider band set by the third. A control of exponent 0.678 reads 0.0282 away, more than seven widths of that band outside it.

The comparison is a measurement, not a proof, calibrated against controls of known exponent. It targets the disputed exponent value; whether it bears on Volkov’s branching model itself is a separate question.

1 Introduction

Kontorovich and Lagarias [1] study the $3x+1$ and $5x+1$ Collatz-type problems through the counting statistics of the associated reverse tree: for a fixed root u , the number $N_u(x)$ of integers up to x that eventually map to u under the accelerated map satisfies $N_u(x) = x^{\eta+o(1)}$ for a conjectural exponent η . Their stochastic model predicts $\eta_{5,BP} \approx 0.650919$ for $q = 5$ (their Theorem 8.10, subscript BP for branching process, in their own notation); a competing branching model due to Volkov, discussed in the same reference, predicts a different value $\eta_{5,BP}^* \approx 0.678$. Kontorovich and Lagarias write that “the empirical data Volkov presents seems insufficient to discriminate between these two predicted exponents,” and separately note “some controversy concerning the conjectured value of the constant.”

2 Setup: the reverse tree of the accelerated $qx + 1$ map

Fix an odd integer $q \geq 3$. The *accelerated* $qx + 1$ map is

$$T_q(n) = \frac{qn + 1}{2^{v_2(qn+1)}}, \quad n \text{ odd},$$

where v_2 is the 2-adic valuation. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node v , adjoining a child $w_a := (2^a v - 1)/q$ for every $a \geq 1$ such that $2^a v \equiv 1 \pmod{q}$; such w_a is automatically an odd integer whenever it is an integer. Writing $d := \text{ord}_q(2)$, admissibility of a for a given parent v depends only on $v \pmod{q}$: if some $a_0(v) \in \{1, \dots, d\}$ satisfies $2^{a_0(v)} v \equiv 1 \pmod{q}$, it is unique, and the admissible exponents are exactly $a \equiv a_0(v) \pmod{d}$; nodes whose residue admits no such a_0 are sterile. At $q = 5$, $d = 4$, every nonzero residue admits an a_0 , and only $v \equiv 0 \pmod{5}$ is sterile.

Every node has exactly one parent under T_q , so a given integer reaches u , if at all, along a single path. We count them:

$$N_u(x) := \#\{n \leq x : n \text{ odd and } T_q^k(n) = u \text{ for some } k \geq 0\}, \quad N_u(x) = x^{\eta+o(1)}.$$

Lemma 2.1 (Sibling congruence). *Let $q \geq 3$ be odd, $d = \text{ord}_q(2)$, and v odd with $2^{a_0} v \equiv 1 \pmod{q}$ solvable. Write the children of v , in order of increasing admissible exponent, as $w_{(j)} := w_{a_0+jd} = (2^{a_0+jd} v - 1)/q$ for $j \geq 0$, where $a_0 = a_0(v)$. Then*

$$w_{(j+1)} = 2^d w_{(j)} + \frac{2^d - 1}{q} \quad (\text{exact identity in } \mathbb{Z}),$$

and consequently $w_{(j+1)} \equiv w_{(j)} + c \pmod{q}$, with $c := ((2^d - 1)/q) \pmod{q}$.

Proof. Direct substitution: $2^d w_{(j)} + (2^d - 1)/q = (2^{a_0+(j+1)d} v - 2^d + 2^d - 1)/q = w_{(j+1)}$. The congruence follows from $2^d \equiv 1 \pmod{q}$. $\square$

When q is prime, $c = 0$ would force $q^2 \mid 2^d - 1$, the base-2 Wieferich condition; the only known Wieferich primes are 1093 and 3511, so $c \neq 0$ for every q of interest here, and consecutive siblings' residues mod q advance by a fixed nonzero step. At $q = 5$, $d = 4$, $c = 3$. Dropping the additive constant, negligible once $w_{(j)}$ is large, leaves the multiplicative step $w_{(j+1)} \approx 2^d w_{(j)}/q$ together with the residue step c ; both extend to any real denominator q_{val} in place of q . §6 uses exactly this to build a branching process with the arithmetic tree's sibling spacing but a tunable exponent.

3 The dispute

Volkov's competing model, as Kontorovich and Lagarias describe it, is a complete binary branching process with a different encoding of the iterates than the arithmetic reverse tree of §2; it predicts $\eta_{5,BP}^* \approx 0.678$, against the $\eta_{5,BP} \approx 0.650919$ of their own stochastic model (their Remark following Theorem 8.10). The two values differ by $\Delta = 0.027$. Kontorovich and Lagarias write that the data Volkov presents cannot settle which of the two is closer to the truth.

4 Method

We enumerate the exact reverse tree of T_5 from 300 roots: odd integers coprime to 5, sampled uniformly below 10^4 and outside every cycle of T_5 , so the entire measurement window lies beyond

them. A single depth-first pass tracks the running path-maximum at every node, giving simultaneous counts at truncation buffers 10^9 through 10^{13} ; this agrees, count for count, with an independent buffer-by-buffer implementation on five held-out roots. A naive confidence interval built from a single fixed buffer excludes every candidate exponent, including the correct one, once the sampling standard error falls below the residual truncation bias; the path-maximum construction sidesteps this by extrapolating the buffer dependence directly. The pooled mean slope across the 300 roots decays geometrically with truncation:

$$0.60049 (10^9) \rightarrow 0.62387 (10^{10}) \rightarrow 0.63261 (10^{11}) \rightarrow 0.63650 (10^{12}) \rightarrow 0.63801 (10^{13}),$$

with increments shrinking by a factor of 0.37 to 0.45 per decade of truncation: the signature of an extrapolable truncation bias.

5 Result: a fixed-window estimator

Aitken’s Δ^2 extrapolation of this sequence to infinite truncation gives

$$\hat{\eta}_5 = 0.639, \quad 95\% \text{ bootstrap interval, fixed window} = [0.633, 0.645]. \quad (1)$$

The bootstrap interval measures variation between sampled roots after the truncation extrapolation; it does not include the remaining bias from the fixed measurement window, and so cannot exclude either asymptotic prediction. That remaining bias is visible directly in a diagnostic panel: on 40 of the 300 roots, at the deepest buffer 10^{13} , the per-decade slope across $10^4 \rightarrow 10^5$ through $10^7 \rightarrow 10^8$ is still rising at the last decade tested, $0.6021 \rightarrow 0.6296 \rightarrow 0.6432 \rightarrow 0.6460$, with no plateau.

Empirical Result 5.1 (Finite-window reading of the counting exponent). *The fixed-window estimator of the $5x + 1$ reverse-tree counting exponent, computed by exact enumeration with Aitken- Δ^2 -extrapolated truncation correction, is 0.639 (bootstrap interval $[0.633, 0.645]$). The sequence of local slopes rises monotonically across the available windows, still increasing at the last one tested; §6 extrapolates the same estimator substantially deeper. On its own, this experiment does not provide a calibrated confidence interval for the asymptotic exponent and does not exclude the Volkov prediction.*

6 A calibrated comparison

The estimator behind Empirical Result 5.1 is biased: on a branching random walk with the arithmetic tree’s own offspring law but an independently drawn branch residue at every node, exponent provably the Kontorovich–Lagarias value 0.650919 itself, realized as a pressure-equation root in a companion paper on the $qx + 1$ branching structure [2], the same window, truncation, and extrapolation read 0.6131. That is an under-read of 0.038, larger than the gap $\Delta = 0.027$ separating the two disputed values.

The bias falls with depth, and depends on how much a process fluctuates, so it can be calibrated out by comparing readings at a common depth. Two constructions share the arithmetic tree’s offspring law and, via Lemma 2.1, its sibling spacing, with a drawn branch residue in place of the endogenous one: one built to exponent 0.650919, one to 0.678. Replacing the integer denominator q in the lemma’s multiplicative step by a tunable real value q_{val} moves the resulting construction’s exponent α to any target solving $q_{\text{val}}^\alpha = q(2^\alpha - 1)$, with q itself held fixed at the arithmetic tree’s own admissibility rule and sterility probability. A third construction, the drawn-residue walk of the paragraph above, shares only the offspring law and draws every branch label independently,

with no sibling congruence; it bounds how much of the calibration depends on that structure. Every construction below runs on the same 300 roots as §4, at checkpoints 10^4 through 10^{10} and truncation buffers 10^9 through 10^{15} , and the arithmetic tree’s reading is checked against them.

Empirical Result 6.1 (Calibrated separation from the Volkov exponent). *At the checkpoint decade $10^9 \rightarrow 10^{10}$, the two congruence-matched constructions of exponent 0.650919 read 0.65122 (drawn residue, sibling congruence) and 0.64981 (its real-valued relaxation), a band 0.00141 wide. The unconstrained construction of the same exponent reads 0.64751, 0.0035 below target; including it widens the band to 0.0037. The arithmetic tree reads 0.64926, below both congruence-matched constructions and inside the wider band set by the unconstrained one. A construction of exponent 0.678 reads 0.67748, 0.0282 away from the arithmetic reading, more than seven widths of that wider band. The same comparison at the shallower decade $10^7 \rightarrow 10^8$, on the same checkpoint grid, gives an arithmetic reading of 0.64622 against 0.67200 for the exponent-0.678 construction, a gap of 0.0258: the separation grows with depth. This measurement targets the exponent value 0.678; Volkov’s branching model itself uses a different encoding of the iterates and is not implemented here.*

Pushed further on the arithmetic tree alone, to checkpoints 10^{12} and buffers 10^{17} , past the point where the drawn-residue constructions develop a heavy tail that stalls their own extrapolation, the slope keeps closing on 0.650919: 0.6465 at $10^7 \rightarrow 10^8$, 0.6487 at $10^8 \rightarrow 10^9$, 0.6490 at $10^9 \rightarrow 10^{10}$, then 0.6506 and 0.6505 at $10^{10} \rightarrow 10^{11}$ and $10^{11} \rightarrow 10^{12}$, the last two within 0.0005 of the prediction. This deeper grid’s 0.6490 at $10^9 \rightarrow 10^{10}$ is close to, but computed on different checkpoint and buffer values than, the 0.64926 of Empirical Result 6.1. The bootstrap bands on these five readings cover root resampling only; a separate truncation term, bounded above by 0.002 (by 0.0003 to 0.0004 from $10^8 \rightarrow 10^9$ onward), is a systematic addressed next.

7 Discussion

Two systematics bound Empirical Result 6.1. The drawn-residue construction and its real-valued relaxation at exponent 0.650919 agree to 0.0035 over six seeds, bounding any implementation systematic between the two recursions at about 0.004, comparable to the wider band width itself. The exponent-0.678 construction fluctuates more than the arithmetic tree and carries a correspondingly larger window-estimator bias, 0.019 against 0.011; that pulls its reading down, toward the arithmetic reading, so the separation survives that direction of bias. Both systematics are smaller than, or comparable to, the 0.0037 wider band width used to read the comparison, and well below the 0.0282 separation itself.

The 0.038 bias of the opening measurement in §6 and the residual bias at the calibration depth, under 0.004 for every construction there, are the same estimator at different depths. The bias falls by roughly an order of magnitude between the window used for the first measurement and the deeper checkpoint decade used for the second.

The comparison settles the narrower question of whether the arithmetic tree’s counting exponent is closer to 0.650919 or to 0.678. It does not implement or test Volkov’s branching model directly, which encodes the iterates differently from the matched constructions built here. The unconstrained construction is Kontorovich and Lagarias’s own stochastic model: accelerating their branching process collapses each run of doublings into a single jump of ratio $2^a/5$ with a drawn uniformly from $\{1, 2, 3, 4\}$, exactly the drawn-label walk used here. The gap between its bias (0.038) and the arithmetic tree’s (0.012 at the same depth) measures how much less the arithmetic tree fluctuates than the model built to describe it, consistent with the arithmetic tree fluctuating less than every

construction in §6. The deep arithmetic-only run sits systematically at or just below 0.650919, a pattern the data here cannot yet distinguish from a slowly converging extrapolation.

Data Availability Statement

Code and data reproducing every claim in this paper, organized by section with a README per subfolder, are at <https://github.com/faculdade/collatz-kl-volkov>.

Acknowledgments

AI assistance (Claude, Anthropic) was used for textual review and translation.

References

- [1] A. V. Kontorovich and J. C. Lagarias, *Stochastic models for the $3x + 1$ and $5x + 1$ problems*, arXiv:0910.1944 (2009); published in: *The Ultimate Challenge: the $3x + 1$ Problem* (J. C. Lagarias, ed.), American Mathematical Society, 2010. Theorem 8.10 and the following Remark are cited from the arXiv version.
- [2] R. A. Tavares, *A closed-form pressure equation for the accelerated $qx + 1$ branching process*, companion paper, in preparation.